Our Solar System: Study Notes

Class 8 Science, Chapter 4

1. The Sun

The Sun is a star at the centre of our solar system. It is about 4.6 billion years old and makes up 99.8% of all the mass in the solar system. Light from the Sun takes about 8 minutes and 20 seconds to reach Earth. The temperature at the Sun's surface is about 5,500 degrees Celsius, while its core reaches about 15 million degrees Celsius.

2. The Eight Planets

There are eight planets. In order from the Sun they are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune. A simple way to remember the order is the sentence: "My Very Educated Mother Just Served Us Noodles."

The first four planets (Mercury, Venus, Earth and Mars) are called rocky or terrestrial planets. The last four (Jupiter, Saturn, Uranus and Neptune) are called gas giants or ice giants.

3. Interesting Facts

- **Mercury** is the smallest planet and the closest to the Sun. A year on Mercury is only 88 Earth days.
- **Venus** is the hottest planet, about 465 degrees Celsius, because its thick clouds of carbon dioxide trap heat. Venus spins backwards compared to most planets.
- **Earth** is the only planet known to have life. About 71% of its surface is covered by water.
- **Mars** is called the Red Planet because of iron oxide (rust) in its soil. It has the tallest volcano in the solar system, Olympus Mons, which is about 22 kilometres high. Mars has two small moons, Phobos and Deimos.
- **Jupiter** is the largest planet. More than 1,300 Earths could fit inside it. Its Great Red Spot is a storm that has lasted for more than 350 years.
- **Saturn** is famous for its rings, which are made mostly of ice and rock. Saturn is so light that it would float in a giant bathtub of water.
- **Uranus** rotates on its side, probably because of a huge collision long ago.
- **Neptune** has the strongest winds in the solar system, up to 2,100 kilometres per hour.

4. Pluto

Pluto was called the ninth planet until 2006. In that year the International Astronomical Union (IAU) reclassified it as a dwarf planet because it has not cleared other objects out of its orbit.

5. Exam Tips

Remember the order of the planets, the difference between rocky planets and gas giants, and why Venus is hotter than Mercury even though Mercury is closer to the Sun.